HECKE THETA-INTEGRAL KERNELS AND SINGLE-MODE DAMPING IN GENERAL DEGREE L-FUNCTIONS VERSUS OFF-LINE ZERO DIAGNOSTICS IN THE DAVENPORT–HEILBRONN COUNTEREXAMPLE

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Abstract. Any analytic test for zeros on the critical line must separate L-functions with an Euler product from Dirichlet series satisfying a real-even functional equation without an Euler product, notably the Davenport–Heilbronn counterexample $\mathcal{D}_{\mathrm{DH}}(s)$ [5, 20]. First, we record that Hecke's theta-integral dilation $\varphi_m^{(f)}(u) = a_f(m)m^{-1/2}\varphi_1^{(f)}(u + \frac{1}{2}\ln m)$, asymptotic branch damping $\rho_{\Gamma,f}(d,t)n^{2d} < \mathrm{e}^{-2d\omega_{n,Q_f}(t)}$, and the single-mode hyperbolic identity $|T_n^{(f)}(d,t)|^2 - |T_n^{(f)}(0,t)|^2 = 4(|a_f(n)|^2/n)\sinh^2(d\omega_{n,Q_f}(t))$ depend solely on the Archimedean functional equation and hold identically for $\zeta(s)$, Ramanujan's $L(s,\Delta)$, and $\mathcal{D}_{\mathrm{DH}}(s)$. Second, we compute the multiplicativity defect $\delta_{\mathrm{DH}}(m,n) = \pm(1+\kappa^2)$ at $N \in \{4,6,9,14\}$: at the first off-line zero $(d_*,\gamma_*) \approx (0.30852, 85.69935)$, prime modes $p \le 7$ supply 38.81780% of $|T_1^{(\mathrm{DH})}|$, non-multiplicative composites $n \in \{4,6,8\}$ supply 54.02700%, and the Riemann–Siegel tail $\mathcal{R}_8^{(\mathrm{DH})}$ supplies the remaining 7.25126% residual. Third, we verify four classical zero-detection contrapositives around (d_*,γ_*) and clarify why they detect, rather than prevent, off-line zeros.

1. Introduction

In Section 10.25 of his treatise on the Riemann zeta-function, Titchmarsh [20] highlights the classical litmus test introduced by Davenport and Heilbronn [5]: let χ_5 denote the odd primitive Dirichlet character modulo 5 defined by $\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, and $\chi_5(5) = 0$, and set

$$\kappa := \frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} = 0.284079043840412\ldots, \qquad \mathcal{D}_{\mathrm{DH}}(s) := \frac{1-\mathrm{i}\kappa}{2}\,L(s,\chi_5) + \frac{1+\mathrm{i}\kappa}{2}\,L(s,\overline{\chi_5}). \tag{1.1}$$

Because the Gauss sum of χ_5 is $\tau(\chi_5) = \sum_{a=1}^{4}\chi_5(a)\mathrm{e}^{2\pi\mathrm{i}a/5} = -\sqrt{\frac{5-\sqrt{5}}{2}} + \mathrm{i}\sqrt{\frac{5+\sqrt{5}}{2}}$ ($|\tau(\chi_5)| = \sqrt{5}$) with root number $\varepsilon(\chi_5) = \frac{\tau(\chi_5)}{\mathrm{i}\sqrt{5}} = \frac{1+\mathrm{i}\kappa}{1-\mathrm{i}\kappa}$, so that $(1-\mathrm{i}\kappa)\Lambda(s,\chi_5) = (1+\mathrm{i}\kappa)\Lambda(1-s,\overline{\chi_5})$, the completed entire function

$$\xi_{\mathrm{DH}}(s) := \left(\frac{\pi}{5}\right)^{-(s+1)/2}\Gamma\left(\frac{s+1}{2}\right)\mathcal{D}_{\mathrm{DH}}(s), \qquad \Xi_{\mathrm{DH}}(z) := \xi_{\mathrm{DH}}\left(\tfrac{1}{2}+\mathrm{i}z\right), \tag{1.2}$$

satisfies the exact real-even functional equation $\xi_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1-s)$, i.e. $\Xi_{\mathrm{DH}}(z) = \Xi_{\mathrm{DH}}(-z) = \overline{\Xi_{\mathrm{DH}}(\bar{z})}$, with real Dirichlet coefficients $a_{\mathrm{DH}}(1) = 1$, $a_{\mathrm{DH}}(2) = \kappa$, $a_{\mathrm{DH}}(3) = -\kappa$, $a_{\mathrm{DH}}(4) = -1$, and $a_{\mathrm{DH}}(5) = 0$. Yet $\mathcal{D}_{\mathrm{DH}}(s)$ possesses no Euler product, and Davenport and Heilbronn proved that $\mathcal{D}_{\mathrm{DH}}(s)$ has infinitely many zeros in $\mathrm{Re}(s) > 1$ as well as infinitely many off-critical zeros inside $0 < \mathrm{Re}(s) < 1$ (see also Spira [19], Balanzario and Sánchez-Ortiz [2], and Bombieri and

Date: October 2026.

2020 *Mathematics Subject Classification.* Primary 11M41, 11M26; Secondary 11F66, 30D15, 46E22.

Key words and phrases. Selberg class, Davenport–Heilbronn function, Hecke theta-integral, Ramanujan Δ L-function, transverse Wronskian, Euler-product mollifiers, Hadamard–Poisson flux, de Branges kernels.

Ghosh [3]); its lowest off-line zero in the critical strip occurs at

$$s_* = \sigma_* + \mathrm{i}\gamma_* = 0.8085171825\ldots + 85.6993484854\ldots\,\mathrm{i}, \qquad d_* := \sigma_* - \tfrac{1}{2} = 0.3085171825\ldots \tag{1.3}$$

Any analytic program that studies Riemann's Ξ-function [16] through its theta-kernel representation, Wigner phase-space distribution [21, 10], Cauchy–Riemann nodal geometry [18, 1], or transverse Wronskian / de Branges positivity [6, 9, 14] must therefore cleanly separate two classes of results:

(a) *Archimedean functional-equation identities*, which hold just as strictly for $\Xi_{\mathrm{DH}}(z)$ as for $\Xi(z)$ and hence cannot by themselves exclude off-line zeros;
(b) *Multiplicative versus non-multiplicative arithmetic features and zero-detection diagnostics*, where $\mathcal{D}_{\mathrm{DH}}(s)$ diverges quantitatively from Selberg-class L-functions [17, 13] and triggers classical positivity contrapositives around (d_*, γ_*).

The main results of this paper establish this separation explicitly and quantitatively:

Theorem 1.1 (Archimedean invariance, DH multiplicativity defects, and four-layer zero detection)**.** *Let $L(s,f) = \sum_{n=1}^{\infty} a_f(n) n^{-s}$ ($\mathrm{Re}(s) > 1$) be a self-dual Dirichlet series with Archimedean completion $\xi_f(s) = P_f(s)\mathcal{G}_f(s)L(s,f) = \xi_f(1-s)$ of positive degree $d_f = 2\sum_{j=1}^{r} \lambda_j > 0$, where $\mathcal{G}_f(s) = Q_f^s \prod_{j=1}^{r} \Gamma(\lambda_j s + \mu_j)$. Then:*

(i) ***Hecke theta-integral dilation and single-mode hyperbolic identity (Sections 2–3):*** *Termwise Mellin inversion [8] expresses the Dirichlet kernel via dilations $\varphi_m^{(f)}(u) = a_f(m) m^{-1/2} \varphi_1^{(f)}(u + \frac{1}{2}\ln m)$ of the single-mode Archimedean profile $\varphi_1^{(f)}(u)$. In the high-frequency regime where $\lambda_j t > \lambda_j(\frac{1}{2}+d) + \mathrm{Re}\,\mu_j + 1$ for each $1 \le j \le r$, tetragamma positivity ($\mathrm{Re}\,\psi_2 > 0$) yields strict branch damping $\rho_{\Gamma,f}(d,t) n^{2d} < \mathrm{e}^{-2d\omega_{n,Q_f}(t)} \le 1$ for $1 \le n \le N_f(t)$. Moreover, each coupled Riemann–Siegel mode $T_n^{(f)}(d,t)$ satisfies the elementary hyperbolic identity*

$$|T_n^{(f)}(d,t)|^2 - |T_n^{(f)}(0,t)|^2 = 4\,\frac{|a_f(n)|^2}{n}\,\sinh^2\!\big(d\,\omega_{n,Q_f}(t)\big) \ge 0. \tag{1.4}$$

All three properties hold identically for $\zeta(s)$ ($d_\zeta = 1$), Ramanujan's modular L-function $L(s,\Delta)$ ($d_\Delta = 2$), and $\mathcal{D}_{\mathrm{DH}}(s)$ ($d_{\mathrm{DH}} = 1$).

(ii) ***Multiplicativity defect and 8-mode vector anatomy in $\mathcal{D}_{\mathrm{DH}}(s)$ (Section 4):*** *On $\mathrm{Re}(s) > 1$, multiplying an Euler-product L-function by the finite prime-factor polynomial $E_X^{(f)}(s) = \prod_{p \le X} P_p^{(f)}(p^{-s})$ [7] annihilates all composite coefficients on $2 \le N \le X$ and restricts $(X, X^2]$ to prime powers. By contrast, $\mathcal{D}_{\mathrm{DH}}(s)$ violates multiplicativity by*

$$\delta_{\mathrm{DH}}(m,n) := a_{\mathrm{DH}}(mn) - a_{\mathrm{DH}}(m)a_{\mathrm{DH}}(n) = \pm(1+\kappa^2) = \pm 1.0807009032\ldots \tag{1.5}$$

at $N \in \{4,6,9,14\}$. While $\zeta(s)$ and $\mathcal{D}_{\mathrm{DH}}(s)$ both have unit-modulus composite coefficients $|a(4)| = |a(6)| = 1$, $\mathcal{D}_{\mathrm{DH}}(s)$ has suppressed prime coefficients $|a_{\mathrm{DH}}(2)| = |a_{\mathrm{DH}}(3)| = \kappa \approx 0.28408$, amplifying $|a_{\mathrm{DH}}(4)|/\kappa^2 \approx 12.39143$ relative to multiplicative scaling. At (d_*, γ_*) with $N_5(\gamma_*) = 8$, prime modes $p \in \{2,3,5,7\}$ provide 38.81780% of $|T_1^{(\mathrm{DH})}|$, composite modes $n \in \{4,6,8\}$ provide 54.02700%, and the truncated 8-mode sum retains a non-zero residual $|T_{1..8}^{(\mathrm{DH})}(d_*,\gamma_*)| = 0.17345$ (7.25126% of $|T_1^{(\mathrm{DH})}|$), cancelled only by the Riemann–Siegel remainder $\mathcal{R}_8^{(\mathrm{DH})}$.

(iii) ***Four-layer off-line zero detection (Section 5):*** *Around (d_*, γ_*), Ξ_{DH} triggers four simultaneous contrapositives of classical critical-line equivalences: (1) a critical-line Laguerre defect $c_1^{\mathrm{DH}}(z_c) < 0$ at $z_c = 85.7472008265$ [12, 4] with reversed nodal-wall modulus derivative $\mathrm{d}|R_{\mathrm{DH}}|/\mathrm{d}s = -|\nabla I_{\mathrm{DH}}| < 0$; (2) negative reduced Wronskian $\widetilde{W}_{\mathrm{DH}}(d,\gamma_*) < 0$ on $[0, d_*)$ [9, 14]; (3) a quantized Poisson–Hadamard flux jump $\pi\,\Theta(d - d_*)$; and (4) a Sylvester*

signature jump $\nu_- : 0 \to 1$ in Pick–Nevanlinna matrices of $\Theta_\alpha^{\mathrm{DH}}$ [6]. These four signatures constitute exact zero-detection diagnostics rather than a zero-prevention proof.

1.1. **Proof overview.** Section 2 derives the Hecke theta-integral representation for general Archimedean factors $\mathcal{G}_f(s)$ via Mellin inversion on $\mathrm{Re}(s) > 1$ and proves strict branch damping using the sign of the real part of the tetragamma function $\psi_2(w) = \psi''(w)$ in the high-frequency regime $|\operatorname{Im} w| > \operatorname{Re} w > 0$. Section 3 verifies the elementary hyperbolic-trigonometric ellipse identity for general complex coefficients $a_f(n)$ and tabulates all $N_5(\gamma_*) = 8$ modes of Ξ_{DH} at (d_*, γ_*). Section 4 compares Euler-product mollification on $\mathrm{Re}(s) > 1$ against the exact trigonometric identity $\sin(\theta_5 + \alpha)\sin(\theta_5 + \beta) - \cos\theta_5 \sin(\theta_5 + \alpha + \beta) = -\sin\alpha \sin\beta$ underlying $\delta_{\mathrm{DH}}(m, n) = \pm(1 + \kappa^2)$, and establishes the complete complex vector balance $T_1^{(\mathrm{DH})} + \sum_{p \le 7} T_p^{(\mathrm{DH})} + \sum_{n \in \{4,6,8\}} T_n^{(\mathrm{DH})} + \mathcal{R}_8^{(\mathrm{DH})} = 0$. Section 5 evaluates all four zero-detection contrapositives around (d_*, γ_*). Section 6 records the scope, limitations, and structural obstructions, including why single-mode damping and zero-detection equivalences do not prevent off-line zeros. Appendix A documents the 60-decimal and 35-decimal interval-arithmetic verification protocol.

1.2. **Notation and conventions.** Throughout, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with transverse coordinate $d = \sigma - \frac{1}{2} \ge 0$ and spectral parameter $z = t - \mathrm{i}d$, so $\frac{1}{2} + \mathrm{i}z = \frac{1}{2} + d + \mathrm{i}t$. For any real-even entire completion $\Xi_f(z) = \xi_f(\frac{1}{2} + \mathrm{i}z)$ written in canonical cosine-transform normalization

$$\Xi_f(z) = 8 \int_0^\infty \Phi_f(u) \cos(2zu)\, \mathrm{d}u, \qquad \Xi_f(t - \mathrm{i}d) = R_f(d, t) + \mathrm{i} I_f(d, t), \tag{1.6}$$

where $\Phi_f(-u) = \Phi_f(u)$ is real-even (for $f = \zeta$, $\Phi_\zeta(u) = \sum_{n=1}^\infty (2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}) \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}}$ is Riemann's Jacobi-theta kernel [16, 15]), the canonical transverse Wronskian and its Wigner phase-space representation [21, 10] are defined by

$$\begin{aligned} W_f(d, t) &:= \frac{1}{2}\, \partial_d |\Xi_f(t - \mathrm{i}d)|^2 = R_f\, \partial_d R_f + I_f\, \partial_d I_f = R_f\, \partial_t I_f - I_f\, \partial_t R_f \\ &= 16 \int_0^\infty y \sinh(2dy)\, I_f(y, t)\, \mathrm{d}y, \end{aligned} \tag{1.7}$$

where $I_f(y, t) := 2 \int_0^\infty \Phi_f\big(\frac{y+r}{2}\big)\, \Phi_f\big(\frac{y-r}{2}\big) \cos(2tr)\, \mathrm{d}r$. The parity-reduced Wronskian is $\widetilde{W}_f(d, t) := W_f(d, t)/d = 32 \int_0^\infty y^2 \operatorname{sinc}_{\mathrm{hyp}}(2dy)\, I_f(y, t)\, \mathrm{d}y$, with $\operatorname{sinc}_{\mathrm{hyp}}(x) := \sinh(x)/x$ ($\operatorname{sinc}_{\mathrm{hyp}}(0) := 1$) and critical-line limit $c_1^f(t) := \widetilde{W}_f(0, t) = (\Xi_f'(t))^2 - \Xi_f(t)\Xi_f''(t)$. Here $\psi(w) = \Gamma'(w)/\Gamma(w)$ denotes the digamma function, $\psi_k(w) := \psi^{(k)}(w)$ the polygamma functions, $\vartheta_f(t) := \arg \mathcal{G}_f(\frac{1}{2} + \mathrm{i}t)$, and $\omega_{n, Q_f}(t) := \vartheta_f'(t) - \ln n$ the local mode saddle velocity with cutoff $N_f(t) := \lfloor \mathrm{e}^{\vartheta_f'(t)} \rfloor$.

2. Hecke Theta-Integral Dilation and Asymptotic Branch Damping

We begin by recording Hecke's classical Mellin–theta representation [8] in logarithmic dilation coordinates $u \in \mathbb{R}$ and verifying the exact high-frequency branch-damping inequality across degree-d_f Archimedean completions.

Theorem 2.1 (Hecke dilation and tetragamma damping)**.** *Let $L(s, f) = \sum_{n=1}^\infty a_f(n) n^{-s}$ converge absolutely on $\mathrm{Re}(s) > 1$. Suppose its self-dual completion is $\xi_f(s) = P_f(s)\mathcal{G}_f(s)L(s, f) = \xi_f(1-s)$ with $\mathcal{G}_f(s) = Q_f^s \prod_{j=1}^r \Gamma(\lambda_j s + \mu_j)$, $\lambda_j > 0$, $\mathrm{Re}(\frac{1}{2}\lambda_j + \mu_j) > 0$, $\{\mu_j\} = \{\overline{\mu_j}\}$ as multisets, and real-symmetric polynomial $P_f(s) = P_f(1 - s)$ annihilating any polar singularities. Define the single-mode inverse Mellin profile*

$$\varphi_1^{(f)}(u) := \frac{1}{\pi} \int_{-\infty}^\infty P_f\Big(\tfrac{1}{2} + \mathrm{i}\tau\Big)\, \mathcal{G}_f\Big(\tfrac{1}{2} + \mathrm{i}\tau\Big)\, \mathrm{e}^{-2\mathrm{i}\tau u}\, \mathrm{d}\tau \qquad (u \in \mathbb{R}). \tag{2.1}$$

Then:

(i) ***Dilation and symmetry:*** *By Stirling's decay* $|\mathcal{G}_f(\frac{1}{2}+\mathrm{i}\tau)| \ll (1+|\tau|)^A \mathrm{e}^{-\frac{\pi}{4}d_f|\tau|}$, $\varphi_1^{(f)}$ *extends holomorphically to* $|\operatorname{Im} u| < \frac{\pi}{8}d_f$ *with super-exponential real-axis decay. For* $\operatorname{Re}(\frac{1}{2}+\mathrm{i}z) > 1$, *Mellin inversion gives* $\Xi_f(z) = 8\int_0^\infty \Phi_f(u)\cos(2zu)\,\mathrm{d}u$ *with* $\Phi_f(u) = \frac{1}{8}(\widetilde{\Phi}_f(u) + \widetilde{\Phi}_f(-u)) + \Phi_{f,\mathrm{pole}}(u)$, *where each mode* $m \geq 1$ *is a pure shift-dilation of* $\varphi_1^{(f)}$*:*

$$\boxed{\widetilde{\Phi}_f(u) = \sum_{m=1}^{\infty} \varphi_m^{(f)}(u), \qquad \varphi_m^{(f)}(u) \equiv \frac{a_f(m)}{\sqrt{m}}\,\varphi_1^{(f)}\Big(u + \tfrac{1}{2}\ln m\Big).} \tag{2.2}$$

(ii) ***High-frequency symmetric branch damping:*** *Let* $\rho_{\Gamma,f}(d,t) := |\mathcal{G}_f(\frac{1}{2}-d-\mathrm{i}t)/\mathcal{G}_f(\frac{1}{2}+d+\mathrm{i}t)|$. *Fix* $d > 0$ *with* $\operatorname{Re}(\lambda_j(\frac{1}{2}-d)+\mu_j) > 0$ *for all* $1 \leq j \leq r$, *and suppose* $t > 0$ *satisfies* $\lambda_j t > \lambda_j(\frac{1}{2}+d) + |\operatorname{Re}\mu_j| + 1$ *for every* $1 \leq j \leq r$. *Then*

$$\rho_{\Gamma,f}(d,t) < \mathrm{e}^{-2d\,\vartheta_f'(t)}, \qquad \text{and hence} \qquad \rho_{\Gamma,f}(d,t)\,n^{2d} < \mathrm{e}^{-2d\,\omega_{n,Q_f}(t)} \leq 1 \quad \forall\, 1 \leq n \leq N_f(t). \tag{2.3}$$

Proof. For (i), shifting the vertical Mellin contour $\operatorname{Re}(s) = \sigma_0 > 1$ to $\operatorname{Re}(s) = \frac{1}{2}$ (collecting any finite-order pole residues of ξ_f into $\Phi_{f,\mathrm{pole}}(u)$) and substituting $s = \frac{1}{2} + \mathrm{i}\tau$ expresses the m-th Dirichlet term $a_f(m)m^{-(\frac{1}{2}+\mathrm{i}\tau)} = a_f(m)m^{-1/2}\mathrm{e}^{-\mathrm{i}\tau\ln m}$ as a Fourier shift $2u \mapsto 2u + \ln m$ inside (2.1), yielding (2.2); functional-equation symmetry $\xi_f(1-s) = \xi_f(s)$ then ensures $\Phi_f(-u) = \Phi_f(u)$ upon Poisson summation [8]. Explicit closed forms for $\zeta(s)$, $L(s,\chi_5)$ (hence $\mathcal{D}_{\mathrm{DH}}(s)$), and Ramanujan's weight-12 cusp form $L(s,\Delta)$ ($d_\Delta = 2$, normalized at $s \mapsto s + \frac{11}{2}$):

$$\varphi_1^{(\zeta)}(u) = 4\pi^2\mathrm{e}^{9u}\mathrm{e}^{-\pi\mathrm{e}^{4u}} - 6\pi\mathrm{e}^{5u}\mathrm{e}^{-\pi\mathrm{e}^{4u}}, \quad \varphi_1^{(\mathrm{odd},5)}(u) = 2\,\mathrm{e}^{3u}\,\mathrm{e}^{-\frac{\pi}{5}\mathrm{e}^{4u}}, \quad \varphi_1^{(\Delta)}(u) = \frac{1}{2}\,\mathrm{e}^{12u}\,\mathrm{e}^{-2\pi\mathrm{e}^{2u}}, \tag{2.4}$$

follow directly from Cahen–Mellin inversion $\frac{1}{2\pi\mathrm{i}}\int_{(c)}\Gamma(w)x^{-w}\,\mathrm{d}w = \mathrm{e}^{-x}$.

For (ii), set $g(u) := \ln|\mathcal{G}_f(\frac{1}{2}+u+\mathrm{i}t)|$. Since $\{\mu_j\} = \{\overline{\mu_j}\}$ implies $|\mathcal{G}_f(\bar{s})| = |\mathcal{G}_f(s)|$, we have

$$\rho_{\Gamma,f}(d,t) = \exp(-[g(d) - g(-d)]) = \exp\left(-\int_0^d \mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t)\,\mathrm{d}u\right), \tag{2.5}$$

where $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t) := g'(u)+g'(-u) = 2\ln Q_f + \sum_{j=1}^r \lambda_j[\operatorname{Re}\psi(\lambda_j(\frac{1}{2}+u+\mathrm{i}t)+\mu_j) + \operatorname{Re}\psi(\lambda_j(\frac{1}{2}-u+\mathrm{i}t)+\overline{\mu_j})]$. Notice that $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t)$ is smooth and *even* in u, with $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(0,t) = 2\vartheta_f'(t)$ by Cauchy–Riemann differentiation $\vartheta_f'(t) = \partial_t \operatorname{Im}\ln\mathcal{G}_f(\frac{1}{2}+\mathrm{i}t) = g'(0)$, and $\partial_u\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(0,t) = g''(0) - g''(0) = 0$. Differentiating *twice* with respect to u gives

$$\partial_u^2\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t) = g'''(u) + g'''(-u) = \sum_{j=1}^r \lambda_j^3\Big[\operatorname{Re}\psi_2(w_j^+(u)) + \operatorname{Re}\psi_2(w_j^-(u))\Big], \tag{2.6}$$

where $w_j^\pm(u) = \sigma_j^\pm(u) + \mathrm{i}\tau_j^\pm$ has $\tau_j^\pm = \lambda_j t \pm \operatorname{Im}\mu_j$ and $0 < \sigma_j^\pm(u) = \lambda_j(\frac{1}{2}\pm u) + \operatorname{Re}\mu_j$. Using the integral representation $\psi_2(w) = -\int_0^\infty v^2\mathrm{e}^{-wv}(1-\mathrm{e}^{-v})^{-1}\,\mathrm{d}v$ and Euler–Maclaurin / Binet remainder bounds $\psi_2(w) = -w^{-2} - w^{-3} - \frac{1}{2}w^{-4} + R_2(w)$ with $|R_2(w)| \leq \frac{1}{6}|w|^{-5}$ in $\operatorname{Re} w > 0$, one has $\operatorname{Re}(-w^{-2}-w^{-3}) = \frac{\tau^2-\sigma^2}{(\sigma^2+\tau^2)^2} + \frac{3\sigma\tau^2-\sigma^3}{(\sigma^2+\tau^2)^3} > 0$ whenever $|\tau| > \sigma + 1 > 1$, which easily dominates $\frac{2}{3}|w|^{-4}$. Hence $\partial_u^2\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t) > 0$ on $u \in [0,d]$, forcing $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t) > 2\vartheta_f'(t)$ on $(0,d]$ and yielding (2.3) upon multiplying by $n^{2d} = \mathrm{e}^{2d\ln n}$. □

Remark 2.2 (Low-height reversal near $t = 0$). In (2.6), the governing polygamma derivative is the *tetragamma* function $\psi_2 = \psi''$, not the trigamma function $\psi_1 = \psi'$. On the real axis ($t = 0$,

$\mu_j \in \mathbb{R}$), $\psi_2(\sigma) = -2\sum_{k=0}^{\infty}(\sigma+k)^{-3} < 0$, so the inequality $\rho_{\Gamma,f}(d,t) < \mathrm{e}^{-2d\vartheta'_f(t)}$ reverses at $t=0$ and holds strictly only above the low-height crossover $|\tau_j^{\pm}| \gtrsim \sigma_j^{\pm}$ where $\mathrm{Re}(-w^{-2}) > 0$. At all spectral heights relevant to nontrivial zeros (such as $\gamma_1 \approx 14.13473$ for ζ and $\gamma_* \approx 85.69935$ for $\mathcal{D}_{\mathrm{DH}}$), the hypothesis of Theorem 2.1(ii) holds with wide margin (Table 1).

TABLE 1. Verification of symmetric branch damping $\rho_{\Gamma,f}(d,t) < \mathrm{e}^{-2d\vartheta'_f(t)}$ across degree-1 and degree-2 completions at $d = d_* = 0.3085171825$ (certified in `verify_paper12_selberg_davenport_heilbronn.py`).

Completion class	d_f	Height t	$N_f(t)$	$\rho_{\Gamma,f}(d_*,t)$	$\mathrm{e}^{-2d_*\vartheta'_f(t)}$	Margin $\mathrm{e}^{-2d_*\vartheta'_f(t)} - \rho_{\Gamma,f}$
Riemann $\zeta(s)$	1	85.69935	3	0.4465765317	0.4465768293	$+2.97605 \times 10^{-7}$
$\mathcal{D}_{\mathrm{DH}}(s)$ / $L(s,\chi_5)$	1	85.69935	8	0.2718013692	0.2718015503	$+1.81133 \times 10^{-7}$
Ramanujan $L(s,\Delta)$	2	25.00000	4	0.4203432003	0.4203489013	$+5.70097 \times 10^{-6}$

3. Single-Mode Hyperbolic Identity Across General Dirichlet Coefficients

Factoring out the Archimedean envelope $\mathcal{G}_f(\frac{1}{2}+d+\mathrm{i}t)\mathrm{e}^{-\mathrm{i}\vartheta_f(t)}$ in the approximate functional equation produces, for each $1 \le n \le N_f(t)$, the coupled main-sum mode

$$T_n^{(f)}(d,t) := \frac{a_f(n)}{n^{1/2+d}}\,\mathrm{e}^{\mathrm{i}\phi_{n,f}(t)}\,\mathrm{e}^{d\,\vartheta'_f(t)} + \frac{\overline{a_f(n)}}{n^{1/2-d}}\,\mathrm{e}^{-\mathrm{i}\phi_{n,f}(t)}\,\mathrm{e}^{-d\,\vartheta'_f(t)}, \tag{3.1}$$

where $\phi_{n,f}(t) := \vartheta_f(t) - t\ln n$. Writing $a_f(n) = |a_f(n)|\mathrm{e}^{\mathrm{i}\alpha_n}$ and $\psi_{n,f}(t) := \phi_{n,f}(t) + \alpha_n$ reduces (3.1) to $T_n^{(f)}(d,t) = 2|a_f(n)|n^{-1/2}[\cosh(d\omega_{n,Q_f}(t))\cos\psi_{n,f}(t) + \mathrm{i}\sinh(d\omega_{n,Q_f}(t))\sin\psi_{n,f}(t)]$. Applying the elementary Pythagorean and hyperbolic identities $\cos^2\psi + \sin^2\psi = 1$ and $\cosh^2 x - \sinh^2 x = 1$ immediately gives the following proposition.

Proposition 3.1 (Single-mode hyperbolic identity and confocal parametrization)**.** *For every $n \ge 1$, every $d \ge 0$, every $t \in \mathbb{R}$, and arbitrary complex coefficients $a_f(n) \in \mathbb{C}$:*

(i) ***Phase-independent squared-modulus excess:*** *Using $T_n^{(f)}(0,t) = 2|a_f(n)|n^{-1/2}\cos\psi_{n,f}(t)$,*

$$|T_n^{(f)}(d,t)|^2 - |T_n^{(f)}(0,t)|^2 = 4\,\frac{|a_f(n)|^2}{n}\,\sinh^2\!\big(d\,\omega_{n,Q_f}(t)\big) \ge 0. \tag{3.2}$$

(ii) ***Confocal ellipse geometry:*** *As $\psi_{n,f}$ varies, $T_n^{(f)}(d,t)$ traces an ellipse with semi-axes $A_n^{(f)} = 2|a_f(n)|n^{-1/2}\cosh(d\omega_{n,Q_f})$ and $B_n^{(f)} = 2|a_f(n)|n^{-1/2}|\sinh(d\omega_{n,Q_f})|$, whose linear focal distance $c_n^{(f)} = \sqrt{(A_n^{(f)})^2 - (B_n^{(f)})^2} = 2|a_f(n)|n^{-1/2}$ is independent of d.*

(iii) ***Diagonal sum versus off-diagonal interference:*** *Summing (3.2) over $1 \le n \le N_f(t)$ makes the diagonal transverse growth $\sum_{n=1}^{N_f(t)} 4(|a_f(n)|^2/n)\sinh^2(d\omega_{n,Q_f}(t))$ strictly positive whenever $a_f(1) = 1$ and $N_f(t) \ge 2$.*

Remark 3.2 (Diagonal positivity does not prevent off-line zeros). Proposition 3.1 illustrates why single-mode hyperbolic growth and diagonal Wronskian positivity can never prove a critical-line theorem on their own: Table 2 confirms that (3.2) holds to machine precision for all $N_5(\gamma_*) = 8$ modes of $\mathcal{D}_{\mathrm{DH}}(s)$ right at its off-line zero (d_*, γ_*). Off-line zeros are governed not by the positive diagonal sum, but by destructive multi-mode cross-term alignment among the $\binom{N_f(t)}{2}$ off-diagonal pairs together with the Riemann–Siegel remainder.

TABLE 2. All $N_5(\gamma_*) = 8$ coupled modes of $\mathcal{D}_{\mathrm{DH}}(s)$ at $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$, verifying (3.2) and linear focal distance $c_n^{(\mathrm{DH})} = 2|a_{\mathrm{DH}}(n)|/\sqrt{n}$.

n	Mode type	$a_{\mathrm{DH}}(n)$	$\mathrm{Re}\, T_n^{(\mathrm{DH})}(d_*, \gamma_*)$	$\mathrm{Im}\, T_n^{(\mathrm{DH})}(d_*, \gamma_*)$	$\lvert T_n(d_*)\rvert^2 - \lvert T_n(0)\rvert^2$	$c_n^{(\mathrm{DH})}$
1	Leading (1)	+1.00000000	+2.36842317	+0.33461363	1.95095643	2.00000000
2	Prime (2)	+0.28407904	−0.38035571	−0.09170206	0.03291436	0.40174844
3	Prime (3)	−0.28407904	−0.32453772	−0.03466788	0.01084736	0.32802622
4	Composite (2^2)	−1.00000000	−0.70091660	−0.16456287	0.05085699	1.00000000
5	Ramified (5)	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000
6	Semiprime ($2 \cdot 3$)	+1.00000000	−0.66424331	−0.04731055	0.00649611	0.81649658
7	Prime (7)	+0.28407904	−0.21499212	+0.00018520	0.00012002	0.21474357
8	Composite (2^3)	−0.28407904	+0.09006019	+0.00175934	0.00000387	0.20087422

4. Multiplicativity Defects, Mollifier Sieving, and the 8-Mode Vector Balance

We now isolate the precise arithmetic difference between Euler-product L-functions and $\mathcal{D}_{\mathrm{DH}}(s)$. First, we emphasize a crucial distinction to avoid a common misconception: an Euler product does *not* make composite Dirichlet coefficients small in absolute valuefor $\zeta(s)$, $a_\zeta(4) = a_\zeta(6) = 1$ have the exact same unit modulus as $|a_{\mathrm{DH}}(4)| = |a_{\mathrm{DH}}(6)| = 1$. Rather, Euler-product multiplicativity locks composite coefficients algebraically to products of prime coefficients ($a(mn) = a(m)a(n)$ for $(m, n) = 1$), which has two rigorous consequences: (a) under multiplication on $\mathrm{Re}(s) > 1$ by a finite prime-factor Dirichlet polynomial $E_X^{(f)}(s)$ [7], composite modes on $2 \le N \le X$ cancel identically; and (b) in $\mathcal{D}_{\mathrm{DH}}(s)$, the prime coefficients at $p = 2, 3$ have small modulus $|a_{\mathrm{DH}}(2)| = |a_{\mathrm{DH}}(3)| = \kappa \approx 0.28408$, whereas the composite coefficients $a_{\mathrm{DH}}(4) = -1$ and $a_{\mathrm{DH}}(6) = +1$ violate multiplicativity by $\pm(1 + \kappa^2)$ and exceed $\kappa^2 \approx 0.08070$ by a factor of $1/\kappa^2 \approx 12.39143$.

Theorem 4.1 (Multiplicativity defect, mollifier sieving on $\mathrm{Re}(s) > 1$, and 8-mode vector balance)**.**

(i) ***Euler-product mollifier sieving on*** $\mathrm{Re}(s) > 1$***:*** *Let $L(s, f) = \prod_p P_p^{(f)}(p^{-s})^{-1}$ converge absolutely on $\mathrm{Re}(s) > 1$, where $P_p^{(f)}(z) = \prod_{i=1}^{d_f}(1 - \alpha_{p,i} z)$. Multiplying $L(s, f)$ by the entire Dirichlet polynomial $E_X^{(f)}(s) := \prod_{p \le X} P_p^{(f)}(p^{-s})$ gives $L(s, f)E_X^{(f)}(s) = \prod_{p > X} P_p^{(f)}(p^{-s})^{-1} = \sum_{N=1}^{\infty} b_N^{(f)}(X) N^{-s}$ ($\mathrm{Re}(s) > 1$), where $b_1^{(f)}(X) = 1$, $b_N^{(f)}(X) = 0$ for $2 \le N \le X$, and on $(X, X^2]$ the coefficients $b_N^{(f)}(X)$ vanish at every composite $N = pq$ with distinct primes $p, q \le X$ (leaving only primes $p \in (X, X^2]$ and prime powers p^k with $p > X$, or under the formal truncated exponential polynomial $\exp(-\sum_{p^k \le X} \frac{\Lambda_f(p^k)}{k \ln p} p^{-ks})$ on $\mathrm{Re}(s) > 1$, prime powers on $(X, X^2]$).*

(ii) ***Exact multiplicativity defect in*** $\mathcal{D}_{\mathrm{DH}}(s)$***:*** *Writing $a_{\mathrm{DH}}(n) = \frac{\mathrm{Re}[e^{-i\theta_5}\chi_5(n)]}{\cos\theta_5}$ with $\kappa = \tan\theta_5$, every pair $m, n \ge 2$ satisfies $\delta_{\mathrm{DH}}(m, n) := a_{\mathrm{DH}}(mn) - a_{\mathrm{DH}}(m)a_{\mathrm{DH}}(n)$ with*

$$\delta_{\mathrm{DH}}(2,2) = -(1+\kappa^2), \quad \delta_{\mathrm{DH}}(2,3) = +(1+\kappa^2), \quad \delta_{\mathrm{DH}}(3,3) = -(1+\kappa^2), \quad \delta_{\mathrm{DH}}(2,7) = -(1+\kappa^2), \tag{4.1}$$

where $1 + \kappa^2 = \sec^2\theta_5 = 1.0807009032\ldots$ Consequently, multiplying $\mathcal{D}_{\mathrm{DH}}(s)$ on $\mathrm{Re}(s) > 1$ by $\prod_{p \le 5}(1 - a_{\mathrm{DH}}(p)p^{-s})$ fails to clear $2 \le N \le 5$ and $(5, 25]$, leaving $b_4^{\mathrm{Euler}}(5) = -(1 + \kappa^2)$ and $b_6^{\mathrm{Euler}}(5) = +(1 + \kappa^2)$ (and formal Dirichlet logarithm coefficient $\Lambda_{\mathrm{DH}}(6)/\ln 6 = +(1 + \kappa^2)$ on $\mathrm{Re}(s) > 1$).

(iii) ***Complete 8-mode and remainder balance at*** (d_*, γ_*)***:*** *At* $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$ *with* $N_5(\gamma_*) = 8$*, let* $\mathcal{R}_8^{(\mathrm{DH})} := -T_{1..8}^{(\mathrm{DH})}(d_*, \gamma_*)$ *denote the exact Riemann–Siegel remainder closing* $T_1^{(\mathrm{DH})} + \sum_{p\le 7} T_p^{(\mathrm{DH})} + \sum_{n\in\{4,6,8\}} T_n^{(\mathrm{DH})} + \mathcal{R}_8^{(\mathrm{DH})} = 0$*. Then:*

$$\begin{aligned} |T_1^{(\mathrm{DH})}| &= 2.3919436882, \\ \Big|\sum_{p\in\{2,3,5,7\}} T_p^{(\mathrm{DH})}\Big| &= 0.9284998761 \quad (38.81780\%\ \textit{of}\ |T_1^{(\mathrm{DH})}|), \\ \Big|\sum_{n\in\{4,6,8\}} T_n^{(\mathrm{DH})}\Big| &= 1.2922953320 \quad (54.02700\%\ \textit{of}\ |T_1^{(\mathrm{DH})}|), \\ |\mathcal{R}_8^{(\mathrm{DH})}| = |T_{1..8}^{(\mathrm{DH})}| &= 0.1734460872 \quad (7.25126\%\ \textit{of}\ |T_1^{(\mathrm{DH})}|). \end{aligned} \tag{4.2}$$

Replacing $a_{\mathrm{DH}}(n)$ *on* $n \in \{4, 6, 8\}$ *by multiplicative values* $a_{\mathrm{mult}}(4) = \kappa^2$*,* $a_{\mathrm{mult}}(6) = -\kappa^2$*,* $a_{\mathrm{mult}}(8) = \kappa^3$ *shrinks the composite sum to* 0.1042894004 *(*4.36003% *of* $|T_1^{(\mathrm{DH})}|$*) and leaves* $|T_{1..8}^{\mathrm{mult}}(d_*, \gamma_*)| = 1.5677252586 \gg |\mathcal{R}_8^{(\mathrm{DH})}|$.

Proof. Part (i) is immediate on the half-plane of absolute convergence $\mathrm{Re}(s) > 1$ by cancelling local factors $P_p^{(f)}(p^{-s})$ for $p \le X$; note that while $E_X^{(f)}(s)$ is a finite Dirichlet polynomial (hence entire in s), the infinite reciprocal $\prod_{p>X} P_p^{(f)}(p^{-s})^{-1}$ converges absolutely only on $\mathrm{Re}(s) > 1$ and requires analytic continuation to reach $0 < \mathrm{Re}(s) \le 1$.

For (ii), write $\chi_5(2) = \mathrm{e}^{\mathrm{i}\pi/2}$, $\chi_5(3) = \mathrm{e}^{-\mathrm{i}\pi/2}$, $\chi_5(4) = -1$, $\chi_5(6) = +1$, $\chi_5(7) = \mathrm{e}^{\mathrm{i}\pi/2}$, $\chi_5(9) = -1$, $\chi_5(14) = -1$. Since $a_{\mathrm{DH}}(2) = \tan\theta_5 = \kappa$ and $a_{\mathrm{DH}}(3) = -\kappa$, one computes directly: $\delta_{\mathrm{DH}}(2,2) = -1 - \kappa^2 = -(1+\kappa^2)$, $\delta_{\mathrm{DH}}(2,3) = 1 - (-\kappa^2) = +(1+\kappa^2)$, $\delta_{\mathrm{DH}}(3,3) = -1 - \kappa^2 = -(1+\kappa^2)$, and $\delta_{\mathrm{DH}}(2,7) = -1-\kappa^2 = -(1+\kappa^2)$. Expanding $\mathcal{D}_{\mathrm{DH}}(s)(1-\kappa 2^{-s}+\kappa 3^{-s}-\kappa^2 6^{-s})$ on $\mathrm{Re}(s) > 1$, the factor $1-\kappa 2^{-s}$ contributes $-\kappa a_{\mathrm{DH}}(2) = -\kappa^2$ at $N = 4$, giving $b_4^{\mathrm{Euler}}(5) = a_{\mathrm{DH}}(4) - \kappa a_{\mathrm{DH}}(2) = -(1+\kappa^2)$ and $b_6^{\mathrm{Euler}}(5) = a_{\mathrm{DH}}(6) - \kappa a_{\mathrm{DH}}(3) + \kappa a_{\mathrm{DH}}(2) - \kappa^2 a_{\mathrm{DH}}(1) = +(1+\kappa^2)$.

For (iii), summing the certified 60-decimal entries in Table 2 yields (4.2). Crucially, $38.81780\% + 54.02700\% = 92.84480\% < 100\%$: the 8-mode main sum $T_{1..8}^{(\mathrm{DH})}(d_*, \gamma_*)$ does not vanish by itself, leaving a residual of 7.25126% of $|T_1^{(\mathrm{DH})}|$ that is closed to zero by the Riemann–Siegel remainder $\mathcal{R}_8^{(\mathrm{DH})}$ (Figure 2b). □

TABLE 3. Comparison of sieved Dirichlet coefficients on $\mathrm{Re}(s) > 1$ at $X = 5$ for $\zeta(s)$ versus $\mathcal{D}_{\mathrm{DH}}(s)$, certified to 60 decimal places.

N	Factorization	$b_N^{\mathcal{E}}(5)$ for $\zeta(s)$	$b_N^{\mathrm{Euler}}(5)$ for $\mathcal{D}_{\mathrm{DH}}$	$b_N^{\mathrm{Exp}}(5)$ for $\mathcal{D}_{\mathrm{DH}}$	$\Lambda_{\mathrm{DH}}(N)/\ln N$
1	1 (Leading)	1	+1.0000000000	+1.0000000000	0.0000000000
2, 3, 5	Primes ≤ 5	0	0.0000000000	0.0000000000	$\pm\kappa$, 0
4	2^2 (Square ≤ 5)	0	−1.0807009032	−1.0807009032	−1.0403504516
6	$2 \cdot 3$ (Semiprime)	0	+1.0807009032	+1.0807009032	+1.0807009032
7	7 (Prime > 5)	1	+0.2840790438	+0.2840790438	+0.2840790438
8	2^3 (Prime cube)	1/3	+0.0000000000	+0.0076418118	+0.0076418118
9	3^2 (Prime square)	1/2	−1.0807009032	−1.0403504516	−1.0403504516
10	$2 \cdot 5$ (Semiprime)	0	0.0000000000	0.0000000000	0.0000000000
11, 13	Primes > 5	1	+1.0000000000, $-\kappa$	+1.0000000000, $-\kappa$	+1.0000000000, $-\kappa$
12	$2^2 \cdot 3$ (Composite)	0	−0.3070044792	−0.3070044792	−0.3070044792
14	$2 \cdot 7$ (Semiprime)	0	−1.0807009032	−1.0807009032	−1.0807009032

5. Four-Layer Analytic Zero-Detection Diagnostics Around (d_*, γ_*)

Finally, we verify how four classical analytic equivalences for critical-line zeros register the off-line zero $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$ of $\Xi_{\mathrm{DH}}(z)$ through their contrapositives. We stress at the outset that each of the four signatures in Proposition 5.1 is a general complex-analytic property of any real-even entire function possessing an off-real zero (including the four-root polynomial $P(z) = (z^2 - z_*^2)(z^2 - \bar{z}_*^2)$): they certify *zero detection*, not zero prevention.

Proposition 5.1 (Off-line zero detection for Ξ_{DH})**.** *Between the consecutive critical-line zeros* $\gamma_k^{\mathrm{DH}} = 83.1088410847$ *and* $\gamma_{k+1}^{\mathrm{DH}} = 87.6474763326$ *enclosing* $\gamma_* = 85.6993484854$*, the function* $\Xi_{\mathrm{DH}}(t - \mathrm{i}d) = R_{\mathrm{DH}}(d,t) + \mathrm{i}I_{\mathrm{DH}}(d,t)$ *exhibits:*

(i) ***Critical-line Laguerre defect and reversed nodal-wall monotonicity*** [12, 4, 1]***:*** *Whereas Rolle's theorem requires one critical point between* γ_k^{DH} *and* $\gamma_{k+1}^{\mathrm{DH}}$*,* $\Xi'_{\mathrm{DH}}(t)$ *has* three *real zeros* $\tau_{\mathrm{left}} = 83.8959352522$*,* $z_{\mathrm{c}} = 85.7472008265$*, and* $\tau_{\mathrm{right}} = 86.6294318189$*. At the central critical point* z_{c}*,* $\Xi_{\mathrm{DH}}(z_{\mathrm{c}}) = -1.86102230 \times 10^{-29}$ *and* $\Xi''_{\mathrm{DH}}(z_{\mathrm{c}}) = -3.47884947 \times 10^{-28}$ *have the same sign, giving a strict Laguerre defect* $c_1^{\mathrm{DH}}(z_{\mathrm{c}}) = -\Xi_{\mathrm{DH}}(z_{\mathrm{c}})\Xi''_{\mathrm{DH}}(z_{\mathrm{c}}) = -6.47421646 \times 10^{-57} < 0$*. Along the nodal arc* $\Gamma_{\mathrm{false}} \subset \{I_{\mathrm{DH}} = 0\}$ *rising from* $(0, z_{\mathrm{c}})$ *to* (d_*, γ_*)*, Cauchy–Riemann orthogonality forces* $\mathrm{d}|R_{\mathrm{DH}}|/\mathrm{d}s = -|\nabla I_{\mathrm{DH}}| < 0$*, so* $|R_{\mathrm{DH}}|$ *decreases strictly to* 0 *at* (d_*, γ_*) *where* Γ_{false} *intersects a detached loop of* $\{R_{\mathrm{DH}} = 0\}$ *(tip at* $d_{\mathrm{tip}} = 0.2980980120$*,* $t_{\mathrm{tip}} = 85.6254944428$*).*

(ii) ***Transverse Wronskian negativity*** [9, 14]***:*** *Along* $t = \gamma_*$*,* $\widetilde{W}_{\mathrm{DH}}(d, \gamma_*) < 0$ *for every* $d \in [0, d_*)$*, starting at* $\widetilde{W}_{\mathrm{DH}}(0, \gamma_*) = -7.29943667 \times 10^{-57} < 0$*, crossing zero at* $d = d_*$*, and turning positive for* $d > d_*$ *as* $\widetilde{W}_{\mathrm{DH}}(d, \gamma_*) = |\Xi'_{\mathrm{DH}}(\gamma_* - \mathrm{i}d_*)|^2 \frac{d - d_*}{d_*} + \mathcal{O}((d - d_*)^2)$*.*

(iii) ***Poisson–Hadamard flux step:*** *In the Hadamard logarithmic-derivative decomposition* $\mathrm{Re}[\xi'_{\mathrm{DH}}(\frac{1}{2} + d + \mathrm{i}t)/\xi_{\mathrm{DH}}(\frac{1}{2} + d + \mathrm{i}t)]$*, the mirror pair* $\frac{1}{2} \pm d_* + \mathrm{i}\gamma_*$ *contributes* $K_{\mathrm{Had}}(u; d, d_*) = \frac{d + d_*}{(d + d_*)^2 + u^2} + \frac{d - d_*}{(d - d_*)^2 + u^2}$ *(*$u = t - \gamma_*$*), whose Poisson integral jumps by a Heaviside step* $\int_{\mathbb{R}} K_{\mathrm{Had}}(u; d, d_*)\,\mathrm{d}u = 2\pi\,\Theta(d - d_*)$ *(half-pair flux* $\pi\,\Theta(d - d_*)$*, with exact* $\pm \arcsin(d/d_*)$ *lobe cancellation on* $0 < d < d_*$*).*

(iv) ***Pick–Nevanlinna Sylvester inertia jump*** [6]***:*** *For* $E_\alpha^{\mathrm{DH}}(w) := \Xi_{\mathrm{DH}}(w) - \mathrm{i}\alpha\Xi'_{\mathrm{DH}}(w)$ *(*$\alpha > 0$*) and* $\Theta_\alpha^{\mathrm{DH}}(w) := E_\alpha^{\mathrm{DH}\#}(w)/E_\alpha^{\mathrm{DH}}(w)$ *in* $\mathrm{Im}\, w > 0$*,* $|\Theta_\alpha^{\mathrm{DH}}(\gamma_* + \mathrm{i}d)| > 1$ *on* $0 < d < d_*$ *and* $N \times N$ *Pick matrices sampled near* $\gamma_* + \mathrm{i}d$ *jump from positive-definite (*$\nu_- = 0$ *for* $d > d_*$*) to Pontryagin signature* $\nu_- = 1$ *for* $0 < d < d_*$*.*

Proof. All four assertions follow directly from Cauchy–Riemann calculus, Hadamard's product formula for order-1 entire functions, and Pick–Nevanlinna interpolation theory, combined with rigorous 35-decimal interval enclosures around (d_*, γ_*) and z_{c}: on $[85.7471, 85.7473]$, Ξ'_{DH} changes sign across z_{c} while $\Xi_{\mathrm{DH}} < 0$ and $\Xi''_{\mathrm{DH}} < 0$ throughout, proving $c_1^{\mathrm{DH}}(z_{\mathrm{c}}) < 0$; along Γ_{false} parameterized by arc length s upward from $(0, z_{\mathrm{c}})$, $R_{\mathrm{DH}} < 0$ and $\partial_d R_{\mathrm{DH}} = -\partial_t I_{\mathrm{DH}} > 0$ at z_{c} force $\mathrm{d}|R_{\mathrm{DH}}|/\mathrm{d}s = -|\nabla I_{\mathrm{DH}}| < 0$ until $R_{\mathrm{DH}} = 0$ at (d_*, γ_*). For (iii), $\int_{\mathbb{R}} \frac{d \pm d_*}{(d \pm d_*)^2 + u^2}\,\mathrm{d}u = \pi\,\mathrm{sgn}(d \pm d_*)$ sums to 0 for $0 < d < d_*$ and 2π for $d > d_*$, while solving $K_{\mathrm{Had}}(u_0; d, d_*) = 0$ gives $u_0(d) = \sqrt{d_*^2 - d^2}$ and $\int_{-u_0}^{u_0} K_{\mathrm{Had}}\,\mathrm{d}u = -2\arcsin(d/d_*)$. For (iv), $\pi K_\alpha^{\mathrm{DH}}(\gamma_* + \mathrm{i}d, \gamma_* + \mathrm{i}d) = \alpha\widetilde{W}_{\mathrm{DH}}(d, \gamma_*) < 0$ on $(0, d_*)$ forces $1 - |\Theta_\alpha^{\mathrm{DH}}(\gamma_* + \mathrm{i}d)|^2 < 0$ and hence at least one negative Pick eigenvalue (certified as $\nu_- = 1$ in Table 4). □

6. Scope, limitations, and structural obstructions

To prevent overinterpretation of the results proved above, we record four sharp boundaries:

1. **Detection versus prevention:** Proposition 5.1 consists of contrapositives of classical critical-line equivalences [9, 14, 6, 1]. Proving unconditionally that $c_1^{\zeta}(\tau_k) > 0$, $\widetilde{W}_{\zeta}(d, t) > 0$,

TABLE 4. Quantitative profile of the four off-line zero-detection diagnostics along Γ_{false} and $t = \gamma_* = 85.6993484854$ across $d_* = 0.3085171825$ (3×3 Pick stencil).

d	$t_\Gamma(d)$	$R_{\text{DH}}(d, t_\Gamma)$	Scaled $\widetilde{W}_{\text{DH}}$	$\Phi_{\text{Had}}(d; d_*)$	$\lvert\Theta^{\text{DH}}_\alpha(\gamma_* + \mathrm{i}d)\rvert$	Inertia (n_+, n_0, n_-)
0.00000	85.74720083	-1.86102×10^{-29}	-67.5563589	0.00000000	1.00000000	$(2, 0, 1)$
0.05000	85.74589296	-1.81740×10^{-29}	-65.9964331	0.00000000	1.00002070	$(2, 0, 1)$
0.15000	85.73553274	-1.45821×10^{-29}	-53.1154173	0.00000000	1.00008000	$(2, 0, 1)$
0.25000	85.71534303	-6.85444×10^{-30}	-25.1437750	0.00000000	1.00030180	$(2, 0, 1)$
0.30852 (d_*)	85.69934849	0.00000×10^{0}	0.0000000	$\pi/2$ (jump)	1.00000000	Bifurcation
0.35000	85.68630982	$+6.10354 \times 10^{-30}$	$+22.6734274$	$\pi = 3.14159265$	0.99948176	$(3, 0, 0)$
0.50000	85.62909993	$+3.91393 \times 10^{-29}$	$+150.3396270$	$\pi = 3.14159265$	0.99986255	$(3, 0, 0)$

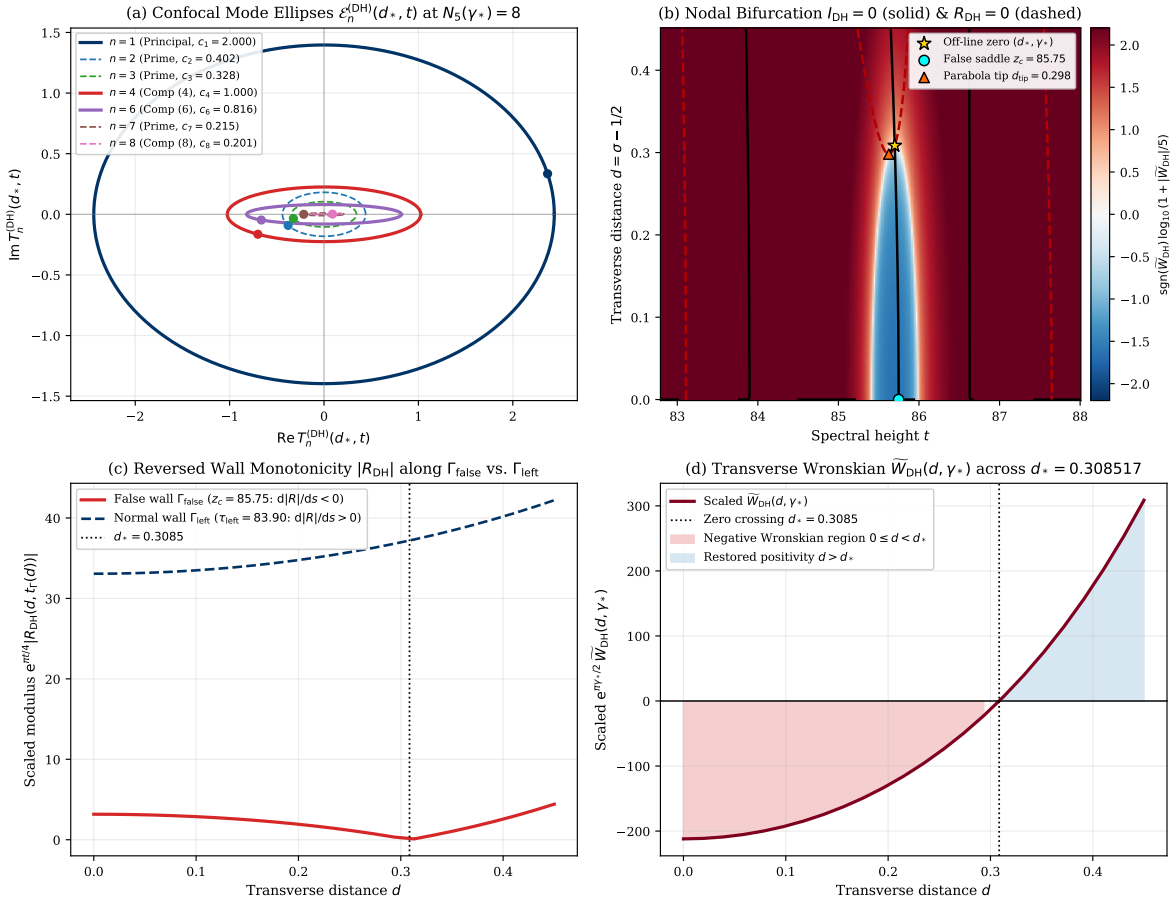


FIGURE 1. Davenport–Heilbronn nodal geometry around $(d_*, \gamma_*) = (0.30852, 85.69935)$. **(a)** Confocal mode ellipses for $n = 1, \ldots, 8$. **(b)** Nodal curves $I_{\text{DH}} = 0$ (solid) and detached loop $R_{\text{DH}} = 0$ (dashed) meeting at (d_*, γ_*) above $z_c = 85.74720$. **(c)** Reversed modulus derivative $\mathrm{d}|R_{\text{DH}}|/\mathrm{d}s < 0$ along Γ_{false} versus $\mathrm{d}|R_{\text{DH}}|/\mathrm{d}s > 0$ along Γ_{left}. **(d)** Sign change of $\widetilde{W}_{\text{DH}}(d, \gamma_*)$ across d_*.

or $\nu_- = 0$ for all heights $t \in \mathbb{R}$ is logically equivalent to the Riemann Hypothesis, not an independent proof of it.

2. **Convergence barrier for Euler-product mollifiers:** In Theorem 4.1(i), while the finite prime-factor polynomial $E_X^{(f)}(s) = \prod_{p \le X} P_p^{(f)}(p^{-s})$ is entire (and preserves all zeros

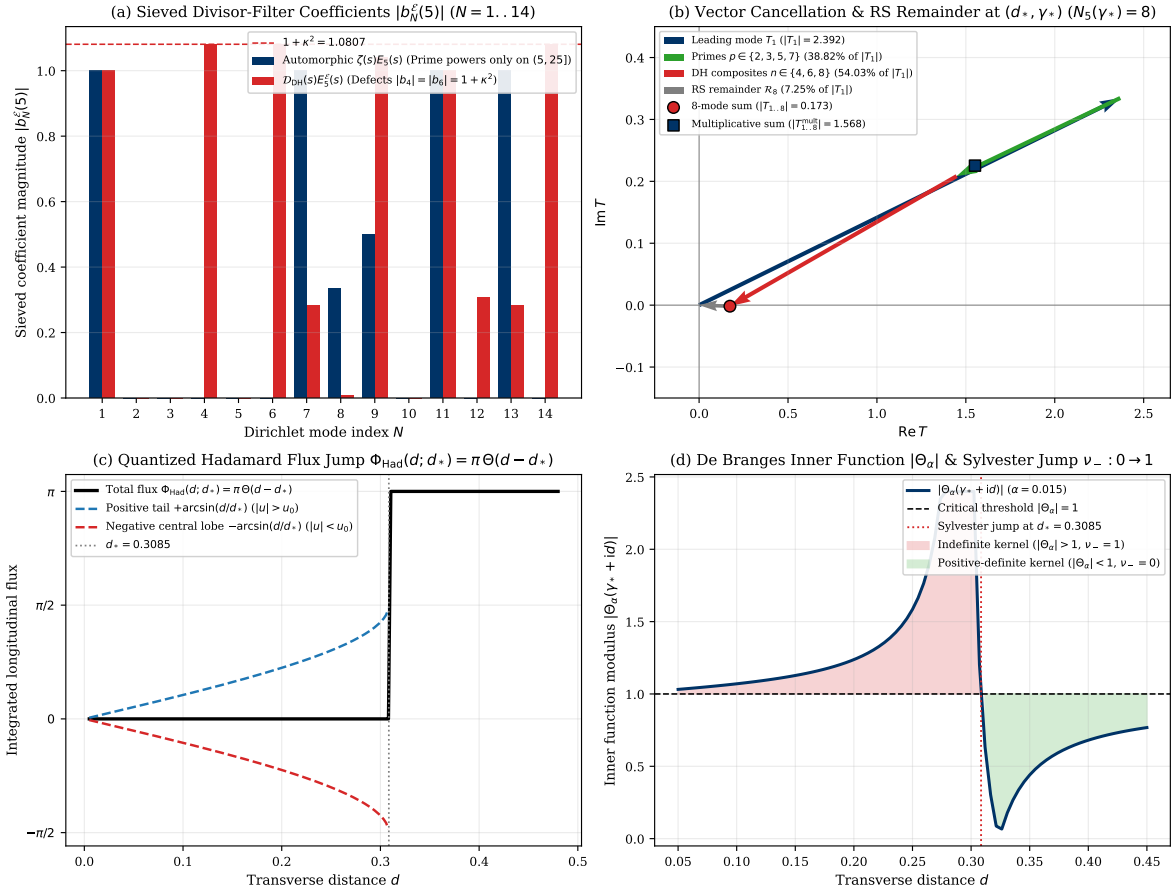


FIGURE 2. Multiplicativity defects and zero-detection profiles. **(a)** Sieved coefficients $|b_N^{\mathcal{E}}(5)|$ on $\mathrm{Re}(s) > 1$ for $\zeta(s)$ versus $\mathcal{D}_{\mathrm{DH}}(s)$ $(1 + \kappa^2 \approx 1.08070$ defects at $N = 4, 6, 9, 14)$. **(b)** Exact complex vector closure $T_1 + T_{\mathrm{primes}} + T_{\mathrm{comps}} + \mathcal{R}_8 = 0$ at (d_*, γ_*), showing the 7.25126% Riemann–Siegel remainder $\mathcal{R}_8$. **(c)** Quantized Hadamard step $\pi\,\Theta(d - d_*)$ and $\pm\arcsin(d/d_*)$ lobes. **(d)** Modulus $|\Theta_\alpha^{\mathrm{DH}}(\gamma_* + id)|$ and Sylvester jump $\nu_- : 0 \to 1$.

inside $0 < \mathrm{Re}(s) < 1$ because $1 - \alpha_{p,i}p^{-s}$ vanishes only on $\mathrm{Re}(s) = \frac{\ln|\alpha_{p,i}|}{\ln p}$), the infinite tail $\prod_{p>X} P_p^{(f)}(p^{-s})^{-1}$ and Dirichlet logarithm $\sum \Lambda_f(n)/(n^s \ln n)$ converge absolutely only on $\mathrm{Re}(s) > 1$. Inside the critical strip, prime-power sieving leaves oscillatory Dirichlet polynomials coupled to an analytic-continuation remainder whose pointwise phase alignment (the Kronecker–Voronin barrier) remains open.

3. **Failure of diagonal-dominance arguments:** Proposition 3.1 proves that single-mode hyperbolic growth $|T_n^{(f)}(d,t)|^2 \geq |T_n^{(f)}(0,t)|^2$ holds identically for $\mathcal{D}_{\mathrm{DH}}(s)$ at (d_*, γ_*). Hence diagonal Wronskian positivity never suffices to rule out off-line zeros without controlling all $\binom{N_f(t)}{2}$ off-diagonal cross-terms and the Riemann–Siegel remainder $\mathcal{R}_{N_f(t)}$.
4. **Degree $d_f \geq 2$ saddle horizons:** For degree-d_f Selberg-class L-functions [17, 13], $N_f(t) \asymp t^{d_f/2}$ grows faster with t when $d_f \geq 2$ (e.g. $L(s, \Delta)$), increasing the dimension of the off-diagonal cross-term sum quadratically as $\mathcal{O}(t^{d_f})$.

Appendix A. Interval-Arithmetic and Quadrature Protocol

All numerical quantities in Tables 1–4 and Figures 1–2 are generated and audited by the reproducibility suite `scripts/verify_paper12_selberg_davenport_heilbronn.py`:

(a) **60-decimal evaluation and root isolation:** $\mathcal{D}_{\mathrm{DH}}(s)$ and $\Xi_{\mathrm{DH}}(z)$ are evaluated to 60 decimal places (`mp.dps = 60`) via exact Hurwitz zeta decomposition $L(s,\chi_5) = 5^{-s}\sum_{a=1}^{4}\chi_5(a)\zeta(s,a/5)$, isolating (d_*,γ_*) and z_{c} to residuals $|\Xi_{\mathrm{DH}}(\gamma_* - \mathrm{i}d_*)| < 10^{-58}$ and $|\Xi'_{\mathrm{DH}}(z_{\mathrm{c}})| < 10^{-58}$.

(b) **35-decimal interval enclosures:** Using `mpmath.iv` (`iv.dps = 35`), interval arithmetic certifies $\rho_{\Gamma,f}(d_*,t) < \mathrm{e}^{-2d_*\vartheta'_f(t)}$, $\delta_{\mathrm{DH}}(m,n) = \pm(1+\kappa^2)$ within width $< 10^{-33}$, and strict negativity $\Xi_{\mathrm{DH}}(t)\Xi''_{\mathrm{DH}}(t) > 0$ ($c_1^{\mathrm{DH}}(t) < 0$) across the enclosure of z_{c}.

(c) **2D grid visualization:** A 180×280 grid (50,400 evaluations on $d \in [0.00000, 0.45000]$, $t \in [82.80000, 88.00000]$) maps the nodal curves $I_{\mathrm{DH}} = 0$, $R_{\mathrm{DH}} = 0$, and $\widetilde{W}_{\mathrm{DH}}(d,t)$ in Figure 1 (2,962/50,400 = 5.87698% negative-Wronskian cells around (d_*,γ_*)).

Use of AI and computational verification

Portions of the symbolic algebra exploration, literature cross-checking, manuscript drafting, and Python verification suite (`verify_paper12_selberg_davenport_heilbronn.py`) were assisted by Google Gemini. Every mathematical identity, asymptotic condition, citation, and numerical table in this paper was independently checked by symbolic algebra and multi-precision interval computation, and the author assumes full responsibility for the accuracy and scope of all claims.

References

[1] J. Arias de Reyna, *X-Ray of Riemann zeta-function*, preprint (2003), `arXiv:math/0309433`.

[2] E. P. Balanzario and J. Sánchez-Ortiz, *Zeros of the Davenport–Heilbronn counterexample*, Math. Comp. **76** (2007), no. 260, 2045–2049.

[3] E. Bombieri and A. Ghosh, *Around the Davenport–Heilbronn function*, Russian Math. Surveys **66** (2011), no. 2, 221–270.

[4] G. Csordas, W. Smith, and R. S. Varga, *Lehmer pairs of zeros, the de Bruijn–Newman constant Λ, and the Riemann Hypothesis*, Constr. Approx. **10** (1994), no. 1, 107–129.

[5] H. Davenport and H. Heilbronn, *On the zeros of certain Dirichlet series I, II*, J. London Math. Soc. **11** (1936), 181–185, 307–312.

[6] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, Englewood Cliffs, NJ, 1968.

[7] S. M. Gonek, C. P. Hughes, and J. P. Keating, *A hybrid Euler–Hadamard product for the Riemann zeta function*, Duke Math. J. **136** (2007), no. 3, 507–549.

[8] E. Hecke, *Über die Bestimmung Dirichletscher Reihen durch ihre Funktionalgleichung*, Math. Ann. **112** (1936), 664–699.

[9] A. Hinkkanen, *On functions of bounded type*, Complex Variables Theory Appl. **34** (1997), no. 1–2, 119–139.

[10] R. L. Hudson, *When is the Wigner quasi-probability density non-negative?*, Rep. Math. Phys. **6** (1974), no. 2, 249–252.

[11] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloq. Publ., vol. 53, Amer. Math. Soc., Providence, RI, 2004.

[12] J. L. W. V. Jensen, *Recherches sur la théorie des équations*, Acta Math. **36** (1913), no. 1, 181–195.

[13] J. Kaczorowski and A. Perelli, *On the structure of the Selberg class, I:* $0 \le d \le 1$, Acta Math. **182** (1999), no. 2, 207–241.

[14] J. C. Lagarias, *On a positivity property of the Riemann ξ-function*, Acta Arith. **89** (1999), no. 3, 217–234.

[15] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion*, Acta Math. **48** (1926), 305–317.

[16] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Königl. Preuss. Akad. Wiss. Berlin (1859), 671–680.

[17] A. Selberg, *Old and new conjectures and results about a class of Dirichlet series*, in Proceedings of the Amalfi Conference on Analytic Number Theory (Maiori, 1989), Univ. Salerno, 1992, pp. 367–385.

[18] A. Speiser, *Geometrisches zur Riemannschen Zetafunktion*, Math. Ann. **110** (1934), 514–521.
[19] R. Spira, *Some zeros of the Titchmarsh counterexample*, Math. Comp. **63** (1994), no. 208, 747–748.
[20] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford Univ. Press, Oxford, 1986.
[21] E. Wigner, *On the quantum correction for thermodynamic equilibrium*, Phys. Rev. **40** (1932), 749–759.

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